Distributions

D.1

Solution:

$$\begin{aligned}
&\frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = -\frac{m_0 g}{\rho}\delta(x-a), && 0 < x < L, t > 0\\
&u(0,t) = u(L,t) = 0, && t > 0\\
&u(x,t) = g(x),\ \frac{\partial u(x,0)}{\partial t} = h(x), && 0 < x < L
\end{aligned}$$

□

D.4

(a) $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$

(b) $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$

Solution:

(a) Here we use $g(x)\delta(x) = g(0)\delta(x)$, and start by differentiating the expression $(g(x)\delta(x))$:

$$(g(x)\delta(x))' = g'(x)\delta(x) + g(x)\delta'(x)$$

$$\downarrow$$

$$g(x)\delta'(x) = (g(x)\delta(x))' - g'(x)\delta(x) =$$

$$= (g(0)\delta(x))' - g'(0)\delta(x) = \boxed{g(0)\delta'(x) - g'(0)\delta(x)}$$

(b) Using the result from (a) and differentiating $(g(x)\delta'(x))$:

$$(g(x)\delta'(x))' = g'(x)\delta'(x) + g(x)\delta''(x)$$
$$\downarrow$$
$$\begin{aligned} g(x)\delta''(x) &= (g(x)\delta'(x))' - g'(x)\delta'(x) = \\ &= (g(0)\delta'(x) - g'(0)\delta(x))' - g'(x)\delta'(x) = \\ &= g(0)\delta''(x) - g'(0)\delta'(x) - g'(0)\delta'(x) + g''(0)\delta(x) = \\ &= \boxed{g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)} \end{aligned}$$

□

D.5

(a) $e^x\delta_1$

(b) $x\delta'$

(c) $\cos(x)\delta'$

(d) $x^2\delta''$

(e) $\int g(t)\delta(t-1)\,\mathrm{d}t$

Solution:

(a)

$$e^x\delta_1 = e^x\delta_1 = \boxed{e\delta_1}$$

(b) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$

$$x\delta' = \boxed{-\delta}$$

(c) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$:

$$\cos(x)\delta' = \cos(0)\delta' + \sin(0)\delta = \boxed{\delta'}$$

(d) Using $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$:

$$x^2\delta'' = \boxed{2\delta}$$

(c) Using $\int g(x)\delta(x-a)\,\mathrm{d}x = g(a)$:

$$\int g(t)\delta(t-1)\,\mathrm{d}t = \boxed{g(1)}$$

□

D.6

(a) $f(x) = e^{-x}\theta(x)$

(b) $f(x) = e^{-|x|}$

Solution:

(a) Using $\theta'(x) = \delta(x)$

$$f'(x) = (e^{-x}\theta(x))' = -e^{-x}\theta(x) + e^{-x}\delta(x) = \boxed{\delta(x) - e^{-x}\theta(x)}$$

$$f''(x) = (\delta(x) - e^{-x}\theta(x))' = \delta'(x) + e^{-x}\theta(x) - e^{-x}\delta(x) = \boxed{\delta'(x) - \delta(x) + e^{-x}\theta(x)}$$

(b) Expressing $|x|$ in distributions:

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases} = x\theta(x) - x(1-\theta(x)) = 2x\theta(x) - x$$

$$f'(x) = (-2\theta(x) - 2x\delta(x) + 1)e^{-|x|} = \boxed{(1-2\theta(x))e^{-|x|}}$$

$$\begin{aligned} f''(x) &= \left(e^{-|x|} - 2e^{-|x|}\theta(x)\right)' = (1-2\theta(x))e^{-|x|} + 2(1-2\theta(x))e^{-|x|}\theta(x) - 2e^{-|x|}\delta(x) = \\ &= e^{-|x|}[(1-2\theta(x)) + 2\theta(x)(1-2\theta(x))] - 2\delta(x) = e^{-|x|}(1-2\theta(x))^2 - 2\delta(x) = \\ &= [(1-2\theta(x))^2 = 1 \text{ for all } x \neq 0] = \boxed{e^{-|x|} - 2\delta(x)} \end{aligned}$$

□

D.8

(b) Determine all primitive distributions for $x\theta(x-1)$

Solution:

(a)

$$\left(x^2\theta(x-1)\right)' = 2x\theta(x-1) + x^2\theta'(x-1) = 2x\theta(x-1) + \delta(x-1)$$

(b) Try to compensate for the $\delta(x-1)$ in (a) by $x^2 \to x^2 - 1$:

$$\left((x^2-1)\theta(x-1)\right)' = 2x\theta(x-1) + (x^2-1)\delta(x-1) = 2x\theta(x-1)$$

Now compensate with a factor 2 and we get the solution

$$\boxed{\left(\frac{x^2-1}{2}\right)\theta(x-1) + C}$$

□

D.9

Solution:

$$\frac{\partial^2 u}{\partial t^2} - S\frac{\partial^2 u}{\partial x^2} = F\delta(x-a)$$
$$u(0,t) = u(1,t) = 0$$

Since we are looking for the stationary solution, $\frac{\partial^2 u}{\partial t^2} = 0$ and we have

$$-\frac{\partial^2 u}{\partial x^2} = \frac{F}{S}\delta(x-a)$$

Integrating both sides twice gives

$$-u(x,t) = \frac{F}{S}(x-a)\theta(x-a) + Ax + B$$

Now check the boundary conditions for this expression:

$$u(0,t) = \underbrace{\frac{F}{S}(-a)\theta(-a)}_{=0 \text{ since } a>0} + B = 0 \to B = 0$$

$$u(1,t) = \frac{F}{S}(a-1)\theta(1-a) + A = 0 \to A = \frac{F}{S}(1-a)$$

$$u(x,t) = \frac{F}{S}((x(1-a) + (a-x)\theta(x-a))$$

□

D.10

Solution:

□

D.12

$$u'(x) + (2x+1)u(x) = \delta'(x)$$

Solution:

Integrating factor for LHS is

$$e^{x^2+x}$$

Multiply this into the equation and integrate LHS, then using the expansion for $f(x)\delta'(x)$:

$$\left(u(x)e^{x^2+x}\right)' = e^{x^2+x}\delta'(x) = e^0\delta'(x) - (2x+1)_{x=0}\delta(x) =$$
$$= \left(u(x)e^{x^2+x}\right)' = \delta'(x) - \delta(x)$$

Now integrating both sides gives:

$$u(x)e^{x^2+x} = \delta(x) - \theta(x) + C$$
$$\downarrow$$
$$\boxed{u(x) = e^{-(x^2+x)}(\delta(x) - \theta(x) + C) = \delta(x) + e^{-x^2-x}(C - \theta(x))}$$

□

D.15

(a) $xU = 0$

(b) $(x-1)U = 0$

(c) $(e^x - 2)U = 0$

(d) $(x^2-1)U = 0$

(e) $\sin(x)U = 0$

(f) $x^3U = 0$

Solution:

(a)

$$U = \delta(x) \rightarrow x\delta(x) = 0 \rightarrow \boxed{U = c\delta(x), \quad c = \text{constant}}$$

(b) Same as in (a) but $\delta(x-1)$

$$\boxed{U = c\delta(x-1)}$$

(c) We can try to set U as $\delta(x-a)$ to make $(e^x - 2) = 0 \rightarrow e^x = 2 \rightarrow x = \ln(2)$

$$\boxed{U = c\delta(x - \ln(2))}$$

(d) Same idea as above, but here we have two roots, $x = 1, x = -1$

$$\boxed{U = c_1\delta(x-1) + c_2\delta(x+1)}$$

(e) Can use $U = \delta(x-a)$ again to make $\sin(x) = 0$

$$\boxed{U = c_k\delta(x - k\pi) = \sum_{k=-\infty}^{\infty} c_k\delta(x - k\pi)}$$

(f) Triple root at $x = 0$, use the relation $x^k U = 0 \rightarrow U = \sum_{k=0}^{n-1} c_k \delta^k(x)$

$$\boxed{U = c_0\delta(x) + c_1\delta'(x) + c_2\delta''(x)}$$

□

D.17

(a) $\theta(-2x+1)$

(b) $\delta(2x)$

(c) $\delta(-2x+1)$

(d) $\delta'(2x)$

Solution:

(a)

$$\theta(-2x+1) = \begin{cases} 1, & -2x+1>0 \to x<\frac{1}{2} \\ 0, & -2x+1<0 \to x>\frac{1}{2} \end{cases}$$

$$\downarrow$$

$$\theta(-2x+1) = \theta\left(-2\left(x-\frac{1}{2}\right)\right) = 1-\theta\left(x-\frac{1}{2}\right) = \boxed{1-\theta_{\frac{1}{2}}(x)}$$

(b) Scaling rule gives

$$\boxed{\delta(2x) = \frac{1}{2}\delta(x)}$$

(c)

$$\delta(-x) = \delta(x), \delta(ax) = \frac{1}{|a|}\delta(x)$$

$$\downarrow$$

$$\delta(-2x+1) = \delta\left(-2\left(x-\frac{1}{2}\right)\right) = \frac{1}{2}\delta\left(x-\frac{1}{2}\right) = \boxed{\frac{1}{2}\delta_{\frac{1}{2}}(x)}$$

(d)

$$\delta'(ax) = \frac{1}{a|a|}\delta'(x)$$

$$\downarrow$$

$$\boxed{\delta'(2x) = \frac{1}{4}\delta'(x)}$$

□

D.18

(a) $\delta * e^{-x^2}$

(b) $\delta' * (e^{-x}\theta(x))$

(c) $\delta_1 * \sin(2x)$

Solution:

Using the definition of the convolution

$$\varphi * U = \int \varphi(x-y)U(y)\,\mathrm{d}$$

(a)

$$\int e^{-(x-y)^2}\delta(y)\,\mathrm{d}y = \boxed{e^{-x^2}}$$

(b)

$$\int e^{-(x-y)}\theta(x-y)\delta'(y)\,\mathrm{d}y = \left[e^{y-x}\theta(x-y)\delta(y)\right]_{-\infty}^{\infty} - \int \left(e^{y-x}\theta(x-y)\right)'\delta(y)\,\mathrm{d}y =$$

$$= \left(e^{-x}\theta(x)\right)' = -e^{-x}\theta(x) + e^{-x}\delta(x) = \boxed{\delta(x) - e^{-x}\theta(x)}$$

(c)

$$\int \sin(2(x-y))\delta(y-1)\,\mathrm{d}y = \boxed{\sin(2x-2)}$$

□

D.22

Solution:

Definition of the cosine series expansion for distributions

$$U = \frac{c_0}{2} + \sum_{k=1}^{\infty} c_k \cos\left(\frac{k\pi x}{L}\right), \quad 0 < x < L$$

$$c_k = \frac{2}{L}\int\limits_0^L U(x)\cos\left(\frac{k\pi x}{L}\right)\mathrm{d}x$$

We will also use the expansion of $f(x)\delta'(x) = f(0)\delta'(x) - f'(0)\delta(x)$.

This gives $(k \neq 0)$

$$c_k = \frac{2}{\pi}\int\limits_0^{\pi} \delta'\left(x - \frac{\pi}{2}\right)\cos(kx)\,\mathrm{d}x = \frac{2}{\pi}\int\limits_0^{\pi} \cos\left(k\frac{\pi}{2}\right)\delta'\left(x - \frac{\pi}{2}\right) + k\sin\left(k\frac{\pi}{2}\right)\delta\left(x - \frac{\pi}{2}\right)\mathrm{d}x =$$

$$= \frac{2}{\pi}\int\limits_0^{\pi} k\sin\left(\frac{k\pi}{2}\right)\mathrm{d}x = \frac{2}{\pi}k\sin\left(\frac{k\pi}{2}\right)$$

and for $k = 0$:

$$c_0 = \frac{2}{\pi}\int_0^\pi \delta'\left(x - \frac{\pi}{2}\right) \mathrm{d}x = \frac{2}{\pi}\left[\delta\left(x - \frac{\pi}{2}\right)\right]_0^\pi = 0$$

$$\boxed{\delta'\left(x - \frac{\pi}{2}\right) = \frac{2}{\pi}\sum_{k=1}^{\infty} k \sin\left(\frac{k\pi}{2}\right)\cos(kx)}$$

□

D.23

Solution:

Whut □

D.24

Solution:

Diffusion,

$$\begin{aligned}
&\frac{\partial u}{\partial t} - D\frac{\partial^2 u}{\partial x^2} = 0, && 0 < x < L, t > 0\\
&\frac{\partial u(0,t)}{\partial x} = \frac{\partial u(L,t)}{\partial t} = 0, && t > 0\\
&u(x,0) = M\delta_{\frac{L}{2}}, && 0 < x < L
\end{aligned}$$

We have homogenous Neumann boundary conditions, which should give us a solution in the form of a cosine series expansion.

$$u(x,t) = \sum u_k(t)\cos\left(\frac{k\pi x}{L}\right)$$

Differentiating termvise and putting into the differential equation gives

$$\sum (u_k'(t) + \alpha u_k(t)) \cos\left(\frac{k\pi x}{L}\right) = 0, \quad \alpha = \frac{Dk^2\pi^2}{L^2}$$
$$\downarrow$$
$$u_k'(t) + \alpha u_k(t) = 0$$
$$\downarrow$$
$$u_k(t) = c_k e^{-\alpha t}$$

We can also expand $u(x,0) = M\delta_{\frac{L}{2}}$ in a cosine series, to get the coefficients, $u_k(0) = c_k$

$$u(x,0) = M\delta_{\frac{L}{2}} = \sum a_k \cos\left(\frac{k\pi x}{L}\right)$$

$$c_0 = \frac{1}{L}\int_0^L M\delta_{\frac{L}{2}}\,\mathrm{d}x = \frac{M}{L}$$

$$c_k = \frac{2}{L}\int_0^L M\delta_{\frac{L}{2}} \cos\left(\frac{k\pi x}{L}\right)\mathrm{d}x = \frac{2M}{L}\cos\left(\frac{k\pi}{2}\right) = , \quad k = 1, 2, 3...$$

This gives the full solution

$$\boxed{u(x,t) = \frac{M}{L}\left(1 + 2\sum_{k=1}^{\infty} \cos\left(\frac{k\pi}{2}\right) e^{-\alpha t} \cos\left(\frac{k\pi x}{L}\right)\right)}$$

□

D.25

Solution:

□

D.27

Solve the three-dimensional potential problem, with $r = \sqrt{x^2 + y^2 + z^2}$

$$\begin{aligned} &\Delta u = \delta(r-2), && 1 < r < 3 \\ &u = 1, && r = 1 \\ &u = 3, && r = 3 \end{aligned}$$

Solution:

Symmetric in r so the Laplacian becomes

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) = \delta(r-2)$$

$$\left(r^2\frac{\partial u}{\partial r}\right)' = r^2\delta(r-2) = 4\delta(r-2)$$

$$\downarrow$$

$$r^2\frac{\partial u}{\partial r} = 4\theta(r-2) + A$$

$$\frac{\partial u}{\partial r} = \frac{4}{r^2}\theta(r-2) + \frac{A}{r^2}$$

$$u(r) = \left(2 - \frac{4}{r}\right)\theta(r-2) - \frac{A}{r} + B$$

$$u(1) = 1 = -\theta(-1) - A + B = 0 \to -A + B = 1$$

$$u(3) = 3 = \left(\frac{2}{3}\right)\theta(1) - \frac{A}{3} + B = \frac{2}{3} + \frac{2}{3}A = 3$$

$$\downarrow$$

$$\begin{cases} -A + B = 1 \\ \frac{2}{3} - \frac{A}{3} + B = 3 \end{cases} \to \frac{2}{3} + \frac{2}{3}A = 2 \to A = 2, B = 3$$

$$\downarrow$$

$$\boxed{u(r) = \left(2 - \frac{4}{r}\right)\theta(r-2) - \frac{2}{r} + 3}$$

□

D.29

Solution:

$$-\Delta u = -u + \delta(x-a)\delta(y-b)\ ,\ \ 0 < x < 2, 0 < y < 1$$

$$-\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = -u + \delta_{a,b}$$

□

D.30

(a) $\delta(t-1)$

(b) $\delta'(t+2)$

(c) 2

(d) $\sin(2t)$

(e) $(t+1)^2$

(f) $e^t\theta(t)$

Solution:

(a) The Fourier transform of $\delta(x)$ is $\mathcal{F}\left[\delta(x)\right] = 1$ and the translation rule is $\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0}\mathcal{F}\left[f(t)\right]$

$$\mathcal{F}\left[\delta(t-1)\right] = e^{-i\omega}$$

(b) The derivative rule is $\mathcal{F}\left[f'(t)\right] = i\omega\mathcal{F}\left[f(t)\right]$. Using this and the traslation rule

$$\mathcal{F}\left[\delta'(t+2)\right] = i\omega e^{2i\omega}$$

(c) Fourier transform of a constant is $2\pi\delta(\omega)$

$$\mathcal{F}\left[2\right] = 4\pi\delta(\omega)$$

(d)

$$\sin(2t) = \frac{e^{2it} - e^{-2it}}{2i}$$

We have $\mathcal{F}\left[e^{i\omega_0 t}f(t)\right] = \mathcal{F}\left[f(\omega-\omega_0)\right]$

$$\mathcal{F}\left[\frac{e^{2it}}{2i}\cdot 1 - \frac{e^{-2it}}{2i}\cdot 1\right] = \frac{\pi}{i}(\delta(\omega-2) - \delta(\omega+2))$$

(e) $\mathcal{F}\left[tf(t)\right] = i\frac{\mathrm{d}}{\mathrm{d}\omega}\mathcal{F}\left[f(t)\right]$

$$\begin{aligned}\mathcal{F}\left[(t+1)^2\right] = \mathcal{F}\left[t^2+2t+1\right] = i^2\frac{\mathrm{d}^2}{\mathrm{d}\omega^2}2\pi\delta(\omega) + i\frac{\mathrm{d}}{\mathrm{d}\omega}4\pi\delta(\omega) + 2\pi\delta(\omega) = \\ = -2\pi\delta''(\omega) + 4\pi i\delta'(\omega) + 2\pi\delta(\omega)\end{aligned}$$

(f) This distribution is not tempered, and thus cannot be Fourier transformed. □

D.31

(a) $\delta(\omega-1)$

(b) $\delta'(\omega+2)$

(c) 2

(d) $\sin(2\omega)$

(e) $(\omega+1)^2$

Solution:

Going backwards on the identities giives

(a)

$$\begin{gathered}
\mathcal{F}\left[1\right] = 2\pi\delta(\omega) \\
\mathcal{F}\left[e^{i\omega_0 t}\cdot 1\right] = 2\pi\delta(\omega-\omega_0) \\
\downarrow \\
\mathcal{F}^{-1}\left[\delta(\omega-1)\right] = \frac{e^{it}}{2\pi}
\end{gathered}$$

(b)

$$\begin{gathered}
\mathcal{F}\left[t\right] = 2\pi i\delta'(\omega) \\
\mathcal{F}\left[e^{i\omega_0 t} t\right] = 2\pi i\delta'(\omega-\omega_0) \\
\downarrow \\
\mathcal{F}^{-1}\left[\delta'(\omega+2)\right] = \frac{e^{-2it}}{2\pi i}t
\end{gathered}$$

(c)

$$\begin{gathered}
\mathcal{F}\left[\delta(t)\right] = 1 \\
\downarrow \\
\mathcal{F}^{-1}\left[2\right] = 2\delta(t)
\end{gathered}$$

(d)

$$\begin{gathered}
\mathcal{F}\left[\theta(t+1)-\theta(t-1)\right] = 2\frac{\sin(\omega)}{\omega} \\
\mathcal{F}\left[\theta\left(\frac{t}{2}+1\right)-\theta\left(\frac{t}{2}-1\right)\right] = \mathcal{F}\left[\theta(t+2)-\theta(t-2)\right] = 2\frac{\sin(2\omega)}{\omega} \\
\mathcal{F}\left[(\theta(t+2)-\theta(t-2))'\right] = \mathcal{F}\left[\delta(t+2)-\delta(t-2)\right] = i\omega\left(2\frac{\sin(2\omega)}{\omega}\right) = 2i\sin(2\omega) \\
\downarrow \\
\mathcal{F}^{-1}\left[\sin(2\omega)\right] = \frac{1}{2i}(\delta(t+2)-\delta(t-2))
\end{gathered}$$

(e)

$$\mathcal{F}\left[\delta'(t)\right] = i\omega$$
$$\mathcal{F}\left[-\delta''(t)\right] = \omega^2$$
$$\mathcal{F}\left[\delta(t)\right] = 1$$
$$\downarrow$$
$$\mathcal{F}^{-1}\left[\omega^2 + 2\omega + 1\right] = -\delta''(t) - 2i\delta'(t) + \delta(t)$$

□

D.34

Solution:

□

D.36

(a) $\delta(t-1)$

(b) $\delta'(t+2)$

(c) 2

(d) $\sin(2t)$

(e) $(t+1)^2$

(f) $e^t\theta(t)$

Solution:

$$\mathcal{L}\left[\delta(t)\right] = 1$$
$$\mathcal{L}\left[f(t-t_0)\right] = e^{-t_0 s}\mathcal{L}\left[f(t)\right]$$
$$\mathcal{L}\left[f'(t)\right] = s\mathcal{L}\left[f(t)\right]$$
$$\mathcal{L}\left[e^{at}f(t)\right] = \mathcal{L}\left[f(t-a)\right]$$
$$\mathcal{L}\left[\theta(t)\right] = \frac{1}{s}$$

(a)

$$\mathcal{L}\left[\delta(t-1)\right] = e^{-s}$$

(b)

$$\mathcal{L}\left[\delta'(t+1)\right] = se^{2s}$$

(c) Not possible

(d) Not possible

(e) Not possible

$$\mathcal{L}\left[(t+1)^2\right] = \mathcal{L}\left[t^2+2t+1\right]$$

(f)

$$\mathcal{L}\left[e^t\theta(t)\right] = \frac{1}{s-1}, \quad \text{Re } s > 0$$

□

D.41

Expand the differential equations to $\mathbb{R}$ so that the boundary conditions are fulfilled.

(a)

$$\begin{cases} u''(x) - iu(x) = 0,\ x > 0 \\ u(0) = 1 \end{cases}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, \quad x > 0 \\ u'(0) = 1 \end{cases}$$

Solution:

Dirichlet conditions give odd expansion, Neumann conditions give even expansions.

(a)

$$\begin{cases} u''(x) - iu(x) = 0, \quad x > 0 \\ u(0) = 1 \end{cases} \to (u^-)'' - iu^- = 2\delta', \quad x \in \mathbb{R}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, \quad x > 0 \\ u'(0) = 1 \end{cases} \to (u^+)'' - iu^+ = 2\delta, \quad x \in \mathbb{R}$$

□
